Dimensional regularisation of the UV–projected row

$\overline{\text{MS}}$ done where it is defined — in momentum space

the $1/r^2$ term Fourier–transformed to the one–loop transverse bubble, then subtracted

0. Conventions

$D = 4 - 2\epsilon$, transverse plane $d \equiv d_\perp = 2 - 2\epsilon$. $\overline{\text{MS}}$ is a *momentum–space* prescription: the loop measure is $\mu^{2\epsilon}\int \frac{d^d\ell}{(2\pi)^d}$ and one subtracts

$$\frac{1}{\hat{\epsilon}} \;\equiv\; \frac{1}{\epsilon} - \gamma_E + \log 4\pi \qquad \Big(\text{equivalently, keep } \tfrac{1}{\epsilon} \text{ and write } \bar{\mu}^2 = 4\pi\mu^2 e^{-\gamma_E}\Big).$$

Kinematics: $\;r = z - x,\quad s = y - x,\quad P = x' - w',\quad q = \bar{\xi}k,\quad D = (r-s)^2 + M^2,\quad M^2 = \frac{\xi}{\bar{\xi}}s^2,\quad D_0 = D|_{r=0} = s^2/\bar{\xi},$ $C_{\rm UV}(\xi) = \xi\bar{\xi} + \frac{\xi}{\bar{\xi}} + \frac{\bar{\xi}}{\xi} = \frac{P_{gg}}{2N_c}$, $\ell_k = \log\frac{k^+}{\Lambda}$.

> **Step 0. $C_{\rm UV}$ carries no $O(\epsilon)$.** The instantaneous δ^{im} vertex supplies an explicit $d_\perp$ and the angular average supplies $1/d_\perp$; they cancel. Nothing multiplying the pole is itself $O(\epsilon)$, so no extra finite term is generated by $\epsilon \times \frac{1}{\epsilon}$.

1. What has to be regulated

$C_{\rm UV}(\xi)$ and $(s\cdot P)/P^2$ do not depend on r, so the whole transverse integration is one object:

$$\mathcal{G} = \int d^2r\; \frac{e^{iq\cdot r}}{r^2\, D}.$$

2. Split the divergence off exactly

$$\frac{1}{r^2 D} = \frac{1}{r^2 D_0} + \frac{1}{r^2}\Big(\frac{1}{D} - \frac{1}{D_0}\Big), \qquad \frac{1}{D} - \frac{1}{D_0} = \frac{2\,r\cdot s - r^2}{D\,D_0},$$

the second bracket vanishing linearly at $r = 0$. So only the first term needs a regulator:

$$\mathcal{G} = \frac{1}{D_0}\,\mathcal{T} \;+\; \frac{1}{D_0}\big[2\,s\cdot\Psi - \Phi\big], \qquad \mathcal{T} \equiv \int d^2r\; \frac{e^{iq\cdot r}}{r^2}.$$

3. $\mathcal{T}$ in momentum space — it is the one–loop transverse bubble

$1/r^2$ is the square of the Weizsäcker–Williams kernel, whose transform is

$$\int d^2r\; e^{i\ell\cdot r}\,\frac{r^i}{r^2} = \frac{2\pi i\,\ell^i}{\ell^2}.$$

By the convolution theorem, with $\frac{1}{r^2} = \frac{r^i}{r^2}\frac{r^i}{r^2}$,

$$\boxed{\mathcal{T} = \int d^2r\; e^{iq\cdot r}\,\frac{1}{r^2} = \int \frac{d^2\ell}{(2\pi)^2}\,\frac{2\pi i\,\ell^i}{\ell^2}\,\frac{2\pi i\,(q-\ell)^i}{(q-\ell)^2} = (2\pi)^2\int \frac{d^2\ell}{(2\pi)^2}\,\frac{\ell\cdot(\ell-q)}{\ell^2(\ell-q)^2}.}$$

The divergence is now where $\overline{\text{MS}}$ expects it: at large ℓ the integrand tends to $1/\ell^2$, logarithmically divergent. There is no infrared problem — at $\ell \to 0$ the numerator $\ell\cdot(\ell-q) \to -\ell\cdot q$ is odd and averages to zero, and likewise at $\ell \to q$.

4. The loop in $d = 2 - 2\epsilon$

Continue the loop measure in the standard way (the two $2\pi i\ell^i/\ell^2$ are vertex factors and stay in two dimensions):

$$\mathcal{T}_\epsilon = (2\pi)^2\,\mu^{2\epsilon}\int \frac{d^d\ell}{(2\pi)^d}\,\frac{\ell\cdot(\ell-q)}{\ell^2(\ell-q)^2}.$$

Reduce with $\ell\cdot(\ell-q)=\ell^2-\ell\cdot q$. The denominator is symmetric under $\ell\to q-\ell$, so $\int\frac{\ell\cdot q}{\ell^2(\ell-q)^2}=\frac{q^2}{2}\int\frac{1}{\ell^2(\ell-q)^2}$, while $\int\frac{d^d\ell}{(\ell-q)^2}$ is scaleless and vanishes:

$$\mathcal{T}_\epsilon=-\frac{q^2}{2}\,(2\pi)^2\,B,\qquad B\equiv\mu^{2\epsilon}\int\frac{d^d\ell}{(2\pi)^d}\,\frac{1}{\ell^2(\ell-q)^2}=\frac{\mu^{2\epsilon}}{(4\pi)^{d/2}}\frac{\Gamma(2-\frac{d}{2})\Gamma(\frac{d}{2}-1)^2}{\Gamma(d-2)}(q^2)^{\frac{d}{2}-2}.$$

At $d=2-2\epsilon$, using $\dfrac{\Gamma(-\epsilon)^2}{\Gamma(-2\epsilon)}=-\dfrac{2}{\epsilon}\dfrac{\Gamma(1-\epsilon)^2}{\Gamma(1-2\epsilon)}$,

$$B=\frac{1}{4\pi q^2}\left(\frac{4\pi\mu^2}{q^2}\right)^{\epsilon}\left(-\frac{2}{\epsilon}\right)c_\Gamma,\qquad c_\Gamma=\frac{\Gamma(1+\epsilon)\Gamma(1-\epsilon)^2}{\Gamma(1-2\epsilon)}=1-\gamma_E\epsilon+O(\epsilon^2),$$

$$\boxed{\mathcal{T}_\epsilon=\frac{\pi}{\epsilon}\,c_\Gamma\left(\frac{4\pi\mu^2}{q^2}\right)^{\epsilon}=\pi\left[\frac{1}{\epsilon}-\gamma_E+\log\frac{4\pi\mu^2}{q^2}\right]+O(\epsilon)=\pi\left[\frac{1}{\hat\epsilon}+\log\frac{\mu^2}{q^2}\right].}$$

5. $\overline{\text{MS}}$

Subtract $\frac{1}{\hat\epsilon}=\frac{1}{\epsilon}-\gamma_E+\log 4\pi$:

$$\mathcal{T}\Big|_{\overline{\text{MS}}}=\pi\,\log\frac{\mu^2}{q^2},\qquad\text{equivalently}\qquad\mathcal{T}_\epsilon=\pi\left[\frac{1}{\epsilon}+\log\frac{\bar\mu^2}{q^2}\right],\quad\bar\mu^2=4\pi\mu^2e^{-\gamma_E}.$$

Why the pole is $+\frac{1}{\epsilon}$ here and $-\frac{1}{\epsilon}$ if you continue the coordinate integral. Continuing $\int d^{2-2\epsilon}r\,e^{iq\cdot r}/r^2$ directly gives $\pi\Gamma(-\epsilon)\left(\frac{q^2}{4\pi\mu^2}\right)^{\epsilon}=-\pi\left[\frac{1}{\epsilon}+\gamma_E+\log\frac{q^2}{4\pi\mu^2}\right]$. The two differ by *exactly* $2\pi/\epsilon$ — a pure pole, no finite part (checked numerically, §9D). The reason is the scaleless integral dropped in §4: $\int\frac{d^d\ell}{(\ell-q)^2}=0=\frac{1}{\epsilon_{\text{UV}}}-\frac{1}{\epsilon_{\text{IR}}}$, which trades the genuine large–ℓ ultraviolet pole for the bubble's small–ℓ infrared one. **The $\overline{\text{MS}}$ finite part — the only thing that enters the answer below — is identical either way:** $\pi\log(\bar\mu^2/q^2)$. The momentum–space form is the one that pairs with the standard $\overline{\text{MS}}$ counterterms, so it is used here.

The finite remainder of §2 is untouched by the continuation:

$$\tilde I(\xi)\equiv\frac{2\,s\cdot\Psi-\Phi}{\pi},\qquad\Phi=2\pi e^{iq\cdot s}K_0(M|q|),$$

$$s\cdot\Psi=\pi\int_0^1 d\alpha\,e^{i\bar\alpha q\cdot s}\left[\bar\alpha s^2\frac{|q|}{\sqrt{\Delta}}K_1(\lambda)+i(q\cdot s)K_0(\lambda)\right],\quad\Delta=\bar\alpha(\alpha s^2+M^2),\ \lambda=|q|\sqrt{\Delta}.$$

6. Back into the row — the $\bar\xi$ cancels

Since $\dfrac{1}{\bar\xi}\cdot\dfrac{1}{D_0}=\dfrac{1}{s^2}$,

$$\boxed{\frac{1}{\bar\xi}\,\frac{s\cdot P}{P^2}\,C_{\text{UV}}(\xi)\,\mathcal{G}=\pi\,C_{\text{UV}}(\xi)\,\frac{(x'-w')\cdot(y-x)}{(x'-w')^2(y-x)^2}\left[\frac{1}{\hat\epsilon}+\log\frac{\mu^2}{\bar\xi^2k^2}+\tilde I(\xi)\right],}$$

using $q=\bar\xi k$. The transverse structure is a product of two WW kernels.

7. The p^+ integral

$$\int_{\Lambda/k^+}^{1-\Lambda/k^+}d\xi\,C_{\text{UV}}=2\ell_k-\frac{11}{6},\qquad\int_{\Lambda/k^+}^{1-\Lambda/k^+}d\xi\,C_{\text{UV}}\log\frac{1}{\bar\xi^2}=\ell_k^2+\frac{\pi^2}{3}-\frac{67}{18},$$

the second being exactly minus Eq. (C.10) of arXiv:1610.03453, since $\log\bar\xi^{-2}=-\log(\xi\bar\xi)+\log(\xi/\bar\xi)$ and the second piece is odd under $\xi\leftrightarrow\bar\xi$ while C_{UV} is even.

8. Final result

Prefactor $\dfrac{1}{(2\pi)^3}\dfrac{g^4}{8\pi^4 k^+}\times\dfrac{\pi}{2\pi}=\dfrac{g^4}{128\pi^7 k^+}=\dfrac{\alpha_s^2}{8\pi^5 k^+}$:

$$\frac{d^3N_2^{\rm NLO}}{d^3k}=\frac{\alpha_s^2(\mu)\,N_c}{8\pi^5 k^+}\int_{x,y,x',w'} e^{-ik\cdot w'}e^{ik\cdot x}\,\frac{(x'-w')\cdot(y-x)}{(x'-w')^2(y-x)^2}\,\Big[\mathcal{P}+\mathcal{F}\Big]$$
$$\times\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[U^{af}(y)-U^{af}(x)\Big]\rho^{b'}(x')\rho^{f}(y),$$

$$\boxed{\mathcal{P}=+\frac{1}{\hat\epsilon}\Big(2\ell_k-\frac{11}{6}\Big)=\Big(\frac{1}{\epsilon}-\gamma_E+\log 4\pi\Big)\Big(2\ell_k-\frac{11}{6}\Big)\ \xrightarrow{\overline{\rm MS}}\ 0}$$

$$\boxed{\mathcal{F}=\Big(2\ell_k-\frac{11}{6}\Big)\log\frac{\mu^2}{k^2}\ +\ \ell_k^2+\frac{\pi^2}{3}-\frac{67}{18}\ +\ \int_{\Lambda/k^+}^{1-\Lambda/k^+}d\xi\,C_{\rm UV}(\xi)\,\tilde I(\xi)}$$

- **The pole is the coupling renormalisation.** $\mathcal{P}\times 2N_c=\frac{1}{\hat\epsilon}\big(4N_c\ell_k-\frac{11N_c}{3}\big)=\frac{1}{\hat\epsilon}\big(4N_c\ell_k-b\big)$ at $N_f=0$: the $\frac{11}{6}$ is the b–function coefficient. In $\overline{\rm MS}$ it is removed together with $-\gamma_E+\log 4\pi$.
- **The scale in the logarithm is k^2, not the dipole size.** After the $\bar\xi$ cancellation the only scale the projected integrand retains is $q=\bar\xi k$, so $\overline{\rm MS}$ delivers $\log(\mu^2/k^2)$ and the $\bar\xi$–dependence becomes $\ell_k^2+\frac{\pi^2}{3}-\frac{67}{18}$. Trading it for $\log\big(\mu^2(y-x)^2\big)$ moves $\log\big(k^2(y-x)^2\big)$ between the two pieces — a rescheming, not done here.
- **$\tilde I$ is regulator–independent.** It is finite in two dimensions, so the continuation does not touch it. Its ξ–integral is not elementary; the endpoints give $e^{ik\cdot(y-x)}\frac{1}{2}\ell_k^2+c_0\ell_k+c_1$.

9. Checks

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A. the d-dimensional transform, formula checked where the answer is known (d=3, a=1)
   pi^(d/2) 2^(d-2a) Gamma((d-2a)/2)/Gamma(a) = 19.739208802179
   known result  int d^3x e^(iqx)/x^2 = 2 pi^2/|q| ,  2 pi^2 = 19.739208802179

B. the epsilon expansion:  pi Gamma(-eps) (q^2/(4 pi mu^2))^eps + pi/eps  ->  pi log(mubar^2/q^2)
   q^2 = 2.7,  mu^2 = 1.3,  mubar^2 = 4 pi mu^2 e^-gamma = 9.17216034208
   eps= 1e-3:  Gamma-form + pi/eps =  3.8369896778214   pi log(mubar^2/q^2) = 3.8419198605062
   eps= 1e-4:  Gamma-form + pi/eps =  3.8414265852134   pi log(mubar^2/q^2) = 3.8419198605062
   eps= 1e-5:  Gamma-form + pi/eps =  3.8418705304043   pi log(mubar^2/q^2) = 3.8419198605062
   eps= 1e-6:  Gamma-form + pi/eps =  3.8419149274703   pi log(mubar^2/q^2) = 3.8419198605062
   eps= 1e-7:  Gamma-form + pi/eps =  3.8419193672023   pi log(mubar^2/q^2) = 3.8419198605062

C. the p+ moments   (lambda = 1e-9, l_k = log(1/lambda))
   int C_UV                = 39.613198342559    2 l_k - 11/6            = 39.613198340559
   int C_UV log(1/xib^2)   = 429.02139290167    l_k^2 + pi^2/3 - 67/18  = 429.02139286022
   int C_UV log(xi/xib)    = -7.6498852e-22  (odd under xi <-> xibar, vanishes)
   int C_UV log(xi xib)    = -429.02139290167   [paper (C.10)];  sum with the row above: -7.6498848e-22

D. MS-bar in momentum space: the 1/r^2 term as the one-loop transverse bubble
   Gamma(-e)^2/Gamma(-2e) = -(2/e) Gamma(1-e)^2/Gamma(1-2e) :
      e=   0.3:      -5.06410115646405      -5.06410115646405   diff -7.889e-31
      e=  0.05:     -39.82284747944915     -39.82284747944915   diff 6.311e-30
      e= -0.02:      99.93608767676722      99.93608767676722   diff -1.262e-29
   the two routes (q^2=2.7, mu^2=1.3):
      e=0.001:  momentum finite part   3.8416741502922   coordinate finite part   3.8369896778214
                  pi  log(mubar^2/q^2)  =  3.8419198605062      difference  of  routes  6283.1899916521   vs  2  pi/e  =
6283.1853071796
      e=0.00001:  momentum finite part   3.8419175126647   coordinate finite part   3.8418705304043
                  pi  log(mubar^2/q^2)  =  3.8419198605062      difference  of  routes  628318.53076494   vs  2  pi/e  =
628318.53071796
      e=0.0000001:  momentum finite part   3.8419198370387   coordinate finite part   3.8419193672023
                  pi  log(mubar^2/q^2)  =  3.8419198605062      difference  of  routes  62831853.071796   vs  2  pi/e  =
62831853.071796
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`zint/src/run_dimreg.py` — the numbers above.